

Choir Divider

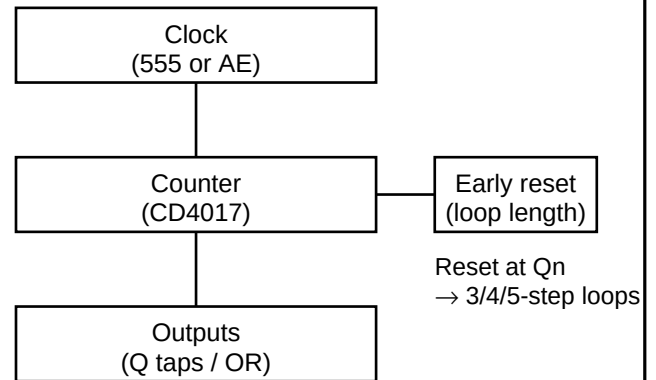
AE Modular BRAEDBOARD • 555 → 4017: counting as rhythm

Core idea: the 4017 counts. Each pulse advances a state (Q0→Q9). Selecting outputs, OR-combining steps, and resetting early turns counting into groove. Optional: use 40106 to clean the clock edges.

Build in layers

- 1) Power + decouple (100nF at each IC).
- 2) Clock: 555 + rate knob (or external AE clock).
- 3) Optional 40106: crisp clock edges.
- 4) 4017: count steps Q0–Q9.
- 5) Divisions: pick Q outputs; reset early for short loops.
- 6) Patterns: diode OR multiple steps.

Concept map



Checkpoints

- Clock present (LED blinks or AE CLK verified).
- 4017 advances one step per pulse.
- RESET returns to Q0 reliably.
- 3+ rhythmic outputs available.
- Double-trigger? add 40106 + shorten CLK wire.

Patch recipes

- Q0 downbeat; Q5 half-time.
- OR(Q0,Q7) accents/masks.
- Reset-at-Q2 = 3-step; reset-at-Q3 = 4-step.
- Swing illusion: Q2/Q3 as late accents.

Build hygiene

Decouple each IC. Keep clock wiring short. Keep RESET pulled low unless you mean it. Use 40106 if you see double-triggers or weird counts.